A Comprehensive Literature Review on Deep Learning Approaches for Medical Image Analysis: Gastric Ulcer Detection, Pneumonia Diagnosis, and Brain Tumor MRI Classification

Daksh Gurjar
Department of Computer Engineering
SVKM's NMIMS, Mukesh Patel School of Technology Management and Engineering, Shirpur - 425405
INDIA

Aditya Yashovardhan
Department of Computer Engineering
SVKM's NMIMS, Mukesh Patel School of Technology Management and Engineering, Shirpur - 425405
INDIA

Jayesh Mandloi
Department of Computer Engineering
SVKM's NMIMS, Mukesh Patel School of Technology Management and Engineering, Shirpur - 425405
INDIA

***Abstract*—This literature review synthesizes findings from sixty peer-reviewed research papers spanning three critical medical imaging domains: gastric ulcer detection from endoscopic images, pneumonia diagnosis from chest X-rays, and brain tumor classification and segmentation from magnetic resonance imaging (MRI). The review examines the methodologies, datasets, performance metrics, and limitations reported across these domains, with particular emphasis on deep learning and artificial intelligence techniques. For gastric ulcer research, convolutional neural networks (CNNs), object detection frameworks such as YOLO variants, and attention mechanisms have demonstrated diagnostic accuracies exceeding 96% in malignancy differentiation tasks. In pneumonia detection, transfer learning with architectures including VGG, ResNet, EfficientNet, and emerging Vision Transformers have achieved classification accuracies ranging from 92% to 99.91%, with recent work addressing explainability and federated learning for privacy preservation. Brain tumor MRI analysis has seen extensive adoption of U-Net variants for segmentation and ensemble models combining ResNet and EfficientNet for classification, with reported accuracies surpassing 99% on benchmark datasets. Across all three domains, common challenges persist: over-reliance on limited public datasets, insufficient external validation, class imbalance, and lack of prospective clinical trials. This review identifies research gaps, compares methodological trends across domains, and provides domain-specific conclusions on the maturity and trajectory of AI-assisted diagnostic research.**

***Index Terms*—Deep learning, medical image analysis, gastric ulcer, pneumonia detection, brain tumor MRI, convolutional neural networks, transfer learning, segmentation, classification.**

I. Introduction

Medical imaging constitutes a cornerstone of modern clinical diagnostics, enabling non-invasive visualization of internal pathologies across virtually every organ system. The convergence of exponential growth in computational power, availability of large-scale annotated datasets, and algorithmic breakthroughs in deep learning has catalyzed a paradigm shift in how medical images are interpreted. Convolutional neural networks (CNNs) and their successors have demonstrated the capacity to match—and in several controlled studies, exceed—the diagnostic performance of trained specialists in tasks ranging from retinal disease grading to pulmonary nodule detection [1]–[3].

This literature review examines sixty research papers organized across three disease domains: (1) gastric ulcer and gastric cancer detection from endoscopic and histopathological images, (2) pneumonia diagnosis from chest X-ray (CXR) and computed tomography (CT) images, and (3) brain tumor classification and segmentation from magnetic resonance imaging (MRI). These three domains were selected for joint analysis because they collectively represent the breadth of AI-driven medical image analysis—spanning gastrointestinal endoscopy, thoracic radiology, and neuro-oncological imaging—while sharing fundamental methodological underpinnings in deep learning architectures, transfer learning strategies, and evaluation paradigms.

The gastric ulcer domain presents unique challenges due to the visual similarity between benign and malignant ulcers in endoscopic images, necessitating sophisticated feature extraction to approximate the diagnostic acumen of experienced gastroenterologists [4], [5]. Pneumonia detection from CXR images has received heightened attention since the COVID-19 pandemic, driving innovation in multi-class classification frameworks that must distinguish viral, bacterial, and COVID-19 pneumonia from normal lung images [7], [8]. Brain tumor MRI analysis encompasses both classification of tumor types (glioma, meningioma, pituitary) and pixel-level segmentation for surgical planning, demanding architectures that balance spatial precision with semantic understanding [3], [6].

The objectives of this review are threefold: (a) to systematically catalog the methodologies, datasets, and performance metrics reported in the sixty source papers; (b) to synthesize findings within each domain, identifying dominant approaches, evolving trends, and persistent gaps; and (c) to draw cross-domain comparisons that illuminate shared challenges and

transferable insights in AI-assisted medical diagnostics.

II. Methodology of the Review

A. Source Material and Verification

The source material for this review comprises exactly sixty research papers deposited in a designated input folder. Prior to analysis, each PDF file was programmatically accessed and its text content extracted using the PyMuPDF library. All sixty files were confirmed accessible and parseable, with no corrupted or unreadable files identified. The papers span publication years from 2018 to 2026, reflecting the rapid evolution of deep learning in medical imaging.

B. Categorization Criteria

Papers were categorized into three domains based on their primary disease focus and imaging modality: (1) Gastric Ulcer/Cancer—papers addressing endoscopic image analysis, histopathological classification, or gastric microbiota profiling (20 papers); (2) Pneumonia—papers addressing chest X-ray or CT-based pneumonia detection, including COVID-19 classification and systematic reviews (20 papers); and (3) Brain Tumor MRI—papers addressing MRI-based brain tumor classification, segmentation, or detection (20 papers). Two papers were identified as potential duplicates (identical title and content appearing under different filenames); these are treated as a single work but both file entries are accounted for in the inventory.

C. Analysis Framework

For each paper, the following elements were extracted: research objective, dataset(s) used (including size and public availability), methodology (model architecture, preprocessing, training strategy), key performance metrics, reported strengths, and acknowledged limitations. The synthesis within each domain section follows a thematic organization—grouping papers by methodological approach rather than presenting individual summaries—to enable comparative analysis and identification of trends.

III. Gastric Ulcer and Gastric Cancer Detection

A. Deep Learning for Endoscopic Image Classification

The application of deep learning to gastric endoscopic image analysis has evolved from binary classification tasks to complex multi-class and multi-task frameworks. Lee *et al.* [9] proposed a computer-aided diagnosis (CADx) system using Xception with depth-wise separable convolutions to classify gastric cancer from pre-cancerous conditions including polyps, ulcers, gastritis, and bleeding. Their preprocessing pipeline incorporated Google's AutoAugment for data augmentation and SLIC superpixel segmentation with FRFCM clustering, achieving an area under the ROC curve (Az) of 0.96 on biopsy-confirmed ground truth.

Hatami *et al.* [10] designed a custom CNN architecture trained on 3,673 endoscopic images of gastric precancerous diseases including erosion, polyps, and ulcers. Using data augmentation, dropout, noise injection, and weight decay for regularization, their model achieved 89% classification accuracy—demonstrating that even moderately sized custom architectures can provide clinically useful diagnostic support, though the accuracy falls below that of deeper transfer learning approaches.

The Gastro-AI model developed by Olimboyev *et al.* [11] leveraged ResNet-50 with transfer learning on the HyperK-vasir dataset—a comprehensive collection of 10,662 endoscopic images spanning 23 disease categories. Trained on 8,529 images and validated on 2,133, the model achieved 90.7% overall accuracy with sensitivity of 89.2% for diseased cases and specificity of 92.1% for healthy tissue. Polyp detection accuracy reached 94.2% and cancer detection 92.7%, though inflammation detection was limited to 87.8% due to image noise.

Kuchkorov *et al.* [12] developed a CNN system for automatic detection of gastric ulcers and lesions in endoscopic images, demonstrating that such systems could process large volumes of stored endoscopic images rapidly with clinically relevant diagnostic ability, potentially reducing endoscopist workload.

Budhiraja and Batra [13] provided a review of deep learning approaches for gastric cancer detection and lesion segmentation, comparing architectures including Mask R-CNN, U-Net, GAS-Net, and GoogLeNet. Their analysis identified GoogLeNet with Inception-v4 and transfer learning as the highest-performing model at 96.7% accuracy.

B. Object Detection and Real-Time Lesion Localization

Object detection frameworks have been adapted for gastric lesion localization, enabling real-time clinical decision support during endoscopic procedures. Ahmad *et al.* [14] enhanced the YOLOv7 algorithm through integration of a Squeeze-and-Excitation (SE) attention block, significantly improving small gastric lesion detection. The attention-powered YOLOv7 achieved precision of 0.72, recall of 0.69, F1-score of 0.71, and mean average precision (mAP) of 0.71 at 63 frames per second—representing a 10% mAP improvement over the baseline YOLOv7.

Tan *et al.* [15] developed a multicenter, retrospective study employing an improved YOLOv8 model with an illumination attention module for real-time instance segmentation and classification of benign versus malignant gastric ulcers. Using a dataset of 9,820 benign, 1,727 malignant, and 15,791 normal mucosa images from four tertiary hospitals in China, the model achieved overall precision, recall, and specificity of 0.91, 0.91, and 0.95 respectively. For malignant ulcer recognition, specificity reached 0.99, demonstrating strong potential for clinical deployment.

Kim *et al.* [16] developed a CNN model for multi-class pathology lesion detection in gastric neoplasms using 10,181 white-light endoscopy images from 2,606 patients. Their six-class categorization (EGC, AGC, dysplasia, BGU, benign polyp, benign erosion) achieved a per-patient lesion detection rate of 95.22%, with 100% detection for EGC and 97.22% for

AGC. Additionally, their T-stage estimation model achieved 85.17% accuracy for distinguishing T1 from T2–T4 cancers.

C. Malignancy Differentiation of Gastric Ulcers

Differentiating malignant from benign gastric ulcers remains a core clinical challenge. Sun *et al.* [4] pioneered a CNN-based gastric ulcer differentiation system using 854 endoscopic images with biopsy-confirmed labels. They compared classification networks against detection network-based approaches that first localized ulcer regions before classification, finding that the detection-based method outperformed endoscopists' diagnostic accuracy.

The same research group advanced this work through non-local attention and densely-connected CNNs [5], incorporating attention blocks into cascaded dense blocks to capture contextual information and enhance pixel-region correlations. On a gastroscopic dataset annotated twice per image by specialists, they achieved 96.79% diagnostic accuracy, 94.92% recall, and an F1-score of 94.70%, with AUC averaging 0.93—approaching gastroenterologist-level performance.

Zheng *et al.* [17] proposed a mechanism mimicking gastroenterologists' diagnostic behavior by integrating segmented malignancy suspicious masks with gastroscopic images. A U-Net network segmented suspicious pathological regions, which then served as attention channels for a ResNet-based classifier. Experiments on a dataset of over 900 patient cases demonstrated substantial performance improvement over standard classification using raw gastroscopic images alone.

Ramamoorthy *et al.* [18] proposed a Medical Diagnostic Decision Support System (MDDSS) framework for early cancer diagnosis and ulcer classification in endoscopy videos, utilizing a Flower Pollination algorithm for color feature selection and achieving 97.56% efficacy for detecting and classifying bleeding cancer and ulcer growth regions across 10,000 EGD datasets.

D. Domain-Specific Pre-Training and Transfer Learning

Lee *et al.* [19] introduced Gastro-BaseNet, a specialized pre-trained model trained exclusively on gastroscopic image data, addressing the domain gap between natural images (ImageNet) and medical endoscopic images. Their comparative study demonstrated that transfer learning from Gastro-BaseNet consistently outperformed ImageNet-based transfer learning for gastric cancer and ulcer prediction, with fine-tuning the entire network yielding the highest performance. The Gastro-BaseNet-based model also exhibited superior lesion localization accuracy.

Yang *et al.* [20] developed a multi-task AI system capable of simultaneously predicting gastric lesions, *Helicobacter pylori* infection status, and lymph node metastasis from endoscopic images, demonstrating the potential for comprehensive diagnostic support from a single AI platform.

Zubair *et al.* [21] presented a framework for gastric cancer classification and segmentation using digital histopathology images, employing an expectation-maximizing Naïve Bayes classifier with a Gaussian Mixture Model. Their framework achieved classification accuracies of 98.87% and 97.28% on two public datasets, with Grad-CAM visualizations enhancing model interpretability.

E. Advanced Classification Approaches

Saba *et al.* [22] introduced a novel approach combining CNN with Q-learning reinforcement for gastrointestinal disease classification and attention-based U-Net for lesion segmentation. The Q-learning component improved learning stability through reward-based feedback, while the attention U-Net highlighted important features for accurate segmentation.

Masmoudi *et al.* [23] presented a framework for automated ulcer classification from wireless capsule endoscopy (WCE) images using the KVASIR dataset. ResNet50 and ResNet152-V2 were trained on preprocessed data, features were fused via concatenation, and optimized using a modified genetic algorithm, achieving 99.67% classification accuracy.

Gnanapriya *et al.* [24] proposed a system for identifying gastric abnormalities with synchronous biopsy site localization using CNN and UNet, providing dual functionality of diagnosis and procedural guidance.

Yoon and Kim [25] reviewed lesion-based CNN approaches for early gastric cancer diagnosis, discussing the design principles for maximizing clinical usefulness when applying CNN to specific organs and diseases.

F. Non-Imaging Clinical Studies

Sari *et al.* [26] conducted a microbiota profiling study of 98 Turkish patients, comparing gastric microbiota composition between non-ulcer dyspepsia (NUD) and peptic ulcer disease (PUD). *H. pylori* DNA was detected in 71.4% of samples, with significantly higher prevalence in PUD patients (82.6% vs. 61.5%). The study identified oral-origin bacteria (*Porphyromonas*, *Neisseria*) that expanded opportunistically in the absence of *H. pylori*, providing microbiological context for AI-based diagnostic research.

G. Trends and Gaps in Gastric Research

The gastric ulcer domain shows a clear trend from simple binary classification toward comprehensive diagnostic systems integrating detection, classification, staging, and even microbiological prediction. Object detection frameworks (YOLO variants) are increasingly preferred for real-time applications. Key gaps include: (1) limited multicenter validation—most studies use single-institution datasets; (2) absence of prospective clinical trials evaluating AI-assisted versus standard endoscopy; (3) underexploration of video-based analysis compared to single-frame approaches; and (4) insufficient integration of clinical metadata (patient history, symptoms) with image-based models.

IV. Pneumonia Detection and Classification

A. CNN-Based Approaches

Convolutional neural networks form the backbone of pneumonia detection research. Sharma and Guleria [27] employed

TABLE I
SUMMARY OF SELECTED STUDIES IN GASTRIC ULCER AND CANCER DETECTION

Study	Architecture / Method	Dataset (Size)	Key Metric (%)
Lee *et al.* [9]	Xception (Depth-wise)	Custom Biopsy-confirmed	Az: 96.0
Olimboyev *et al.* [11]	ResNet-50	HyperKvasir (10,662)	Acc: 90.7
Ahmad *et al.* [14]	YOLOv7 + SE Attention	Custom Endoscopic	mAP: 71.0
Tan *et al.* [15]	YOLOv8 + Attention	Multicenter (27,338)	Spec: 99.0
Sun *et al.* [5]	Densely-connected CNN	Custom Gastroscopic	Acc: 96.79
Masmoudi *et al.* [23]	ResNet50 + ResNet152-V2	KVASIR WCE	Acc: 99.67

VGG16 combined with neural networks for pneumonia detection on two CXR datasets, achieving 92.15% accuracy on a standard dataset and 95.4% on a three-class dataset (pneumonia, normal, COVID-19). Their comparative analysis showed VGG16+NN outperforming VGG16 combined with SVM, KNN, Random Forest, and Naïve Bayes classifiers.

Szepesi and Szilágyi [28] proposed a novel CNN architecture featuring dropout in the convolutional layers—an unconventional design choice that improved regularization. Trained on 5,856 labeled CXR images from pediatric patients, their network achieved 97.2% accuracy, 97.3% recall, 97.4% precision, and AUC of 0.982, which would have placed first in the corresponding Kaggle competition.

Potharaju *et al.* [29] integrated Convolutional Block Attention Module (CBAM) and Squeeze-and-Excitation (SE) networks into CNNs for pneumonia detection from 5,816 X-rays. The CBAM-enhanced CNN achieved 98.6% accuracy with 98.3% sensitivity and 97.9% specificity, while the SE-integrated CNN reached 96.25%. A lightweight web application was developed for deployment in low-resource healthcare settings.

B. Transfer Learning and Pre-Trained Models

Transfer learning has become the predominant approach for pneumonia classification. Ali *et al.* [30] evaluated six deep learning models on 5,856 CXR images, finding EfficientNetV2L achieved the highest accuracy at 94.02%, outperforming CNN (87.78%), InceptionResNetV2 (88.94%), Xception (90.7%), VGG16 (91.66%), and ResNet50 (87.98%).

Shilpa *et al.* [31] compared ResNet50, MobileNetV2, AlexNet, EfficientNetB0, and Xception for pneumonia detection, incorporating CLAHE preprocessing and cross-validation. EfficientNetB0 achieved the highest test accuracy of 99.78% with perfect precision, recall, and F1 scores, demonstrating the importance of data preprocessing in achieving high performance.

Lamouadene *et al.* [32] applied transfer learning using ResNet18 combined with SVM on 21,165 CXR images, achieving 98% accuracy with SGD optimization. Additional models using EfficientNet-CNN and Xception-CNN achieved 99.20% and 98.80% accuracy respectively.

Agrawal and Gandhi [33] introduced PneumoSense++, a framework comparing CNN, DenseNet121, and ResNet50 across seven optimizers (Adam, SGD, AdaGrad, AdaMax, RMSprop, Nadam, AdaDelta) on 28,819 X-ray images. DenseNet121 with SGD achieved the highest performance: 94.80% accuracy, 97.69% recall, and 98.86% AUC.

C. Hybrid and Ensemble Methods

Hybrid approaches combining multiple architectures or feature extraction strategies have shown promise. Akshay *et al.* [34] proposed a dual-CNN feature fusion strategy combining VGG16 and ResNet features, applying PCA for dimensionality reduction retaining 95% variance, and classifying using SVM and Random Forest—replacing the typical CNN softmax with more interpretable ML classifiers.

Munir *et al.* [35] developed PneuX-Net, an ensemble framework integrating Random Forest, Gaussian Naïve Bayes, and K-Nearest Centroid. Using 10-fold cross-validation, the PneuX-Net+KNC combination achieved 99.91% accuracy, representing one of the highest reported figures in pneumonia classification.

Hashmi *et al.* [36] presented a hybrid approach combining ResNet-50 deep features with 2D-discrete wavelet transform (DWT) and Gray-Level Co-occurrence Matrix (GLCM) handcrafted features, demonstrating that fusion of learned and engineered features can enhance classification robustness.

D. Vision Transformers and Attention Mechanisms

The introduction of transformer architectures to medical imaging represents a significant methodological shift. The DAViT model [37] proposed a Domain-Adapted Vision Transformer combining ViT self-attention for global feature capture with shallow CNNs for local feature extraction. Evaluated on 5,856 CXR images, DAViT achieved a 97% F1-score and 96% AUC for pneumonia detection, outperforming twelve baseline methods. For pneumonia type classification (bacterial vs. viral), DAViT achieved 81% F1-score—significantly surpassing baselines by 25–74%. Grad-CAM heatmaps provided interpretable visualizations.

Kunjumon *et al.* [38] developed a hybrid model utilizing Swin Transformer's self-attention and cross-attention mechanisms for hierarchical feature extraction, combined with CNN-based classification. Residual connections and attention mechanisms enhanced gradient flow and model convergence.

E. Federated Learning and Privacy-Preserving Methods

Rana and Marwaha [39] addressed data privacy through a federated learning approach using autoencoder neural networks trained on decentralized CXR datasets. This unsupervised learning paradigm enables multiple healthcare institutions to collaboratively develop diagnostic models without

sharing raw patient data—a critical consideration for real-world deployment.

F. Explainability and Interpretability

Ren *et al.* [40] proposed an interpretable pneumonia detection system combining neural networks with Bayesian Networks on a dataset of 35,389 community-acquired pneumonia cases. Their multi-source data approach improved performance while providing intuitive explanations for diagnostic results, addressing the critical "black box" limitation of deep learning in clinical settings.

Li and Li [7] developed an AI deep learning platform using voting algorithms combining multiple CNNs for COVID-19 pneumonia detection from 10,182 CXR images, achieving 99.95% diagnostic accuracy for COVID-19 among viral pneumonias.

G. Systematic Reviews and Comparative Studies

Khan *et al.* [1] conducted a systematic literature review of intelligent pneumonia identification from CXR, analyzing algorithm usability, computational complexity, and dataset quality. Their findings indicated that deep learning algorithms achieve 98.7% accuracy with 0.99 sensitivity and 0.98 specificity, while highlighting that the majority of applied datasets are highly imbalanced and limited.

Rickard *et al.* [41] performed a scoping review of 35 studies on ML-based viral versus bacterial pneumonia differentiation in pediatric populations. Key findings include: 31 of 35 studies used the Kermany dataset, raising overfitting concerns; median binary classification accuracy was 92.3% (range: 80.8–97.9%); and median multiclass accuracy was 91.8% (range: 76.8–99.7%).

Aljuaid *et al.* [8] conducted a rigorous experimental comparison of six CNN architectures (ResNet, AlexNet, VGGNet, SqueezeNet, DenseNet, InceptionV3) using a uniform pipeline on 6,939 balanced CXR images. VGGNet achieved the highest overall accuracy of 97%.

Hasan *et al.* [42] provided a comprehensive summary of DL-based pneumonia diagnosis systems, analyzing datasets and training methodologies across the field.

H. Trends and Gaps in Pneumonia Research

The pneumonia domain is characterized by: (1) overwhelming reliance on the Kermany/Guangzhou pediatric CXR dataset, limiting generalizability to adult populations and diverse imaging conditions; (2) a trend toward hybrid architectures combining CNNs with transformers or traditional ML classifiers; (3) growing emphasis on explainability through Grad-CAM and Bayesian approaches; and (4) emerging interest in privacy-preserving federated learning. Critical gaps include the need for: multi-institutional prospective validation; standardized benchmarking protocols enabling fair model comparison; exploration of CT-based pneumonia detection (most studies use CXR); and integration of clinical variables beyond imaging data.

V. Brain Tumor MRI Classification and Segmentation

A. Transfer Learning for Tumor Classification

Transfer learning from ImageNet-pretrained models dominates brain tumor classification research. Noreen *et al.* [6] proposed multi-level feature extraction and concatenation using Inception-v3 and DenseNet201, extracting features from different inception modules and dense blocks before concatenation. On a publicly available three-class dataset, the method achieved 99.34% (Inception-v3) and 99.51% (DenseNet201) testing accuracy.

Younis *et al.* [43] developed an enhanced ResNet50 model with data augmentation and regularization for classifying meningioma, glioma, pituitary, and no-tumor classes using 5,712 images. Class-specific accuracies reached 98.33% (glioma), 94.44% (meningioma), 100% (no tumor), and 100% (pituitary), with overall AUC of 0.999.

Asif *et al.* [46] conducted a systematic comparison of Xception, NasNet Large, DenseNet121, and InceptionResNetV2 with three optimizers (ADAM, SGD, RMSprop) on two benchmark datasets. Their comprehensive analysis provided guidance for selecting optimal architecture-optimizer combinations.

Ahmad and Choudhury [47] evaluated seven transfer learning methods (VGG-16, VGG-19, ResNet50, InceptionResNetV2, InceptionV3, Xception, DenseNet201) each followed by five classifiers (SVM, Random Forest, Decision Tree, AdaBoost, Gradient Boosting), achieving a best accuracy of 99.39% with 10-fold cross-validation.

Rizwan *et al.* [48] proposed a Gaussian Convolutional Neural Network (GCNN) evaluated on two datasets—one for three-class tumor classification (3,064 images from 233 patients) and another for glioma grade classification (516 images from 73 patients)—achieving 99.8% and 97.14% accuracy respectively.

Rehman *et al.* [49] tested LeNET, MobileNetV2, DenseNet, ResNet, and EfficientNet for brain tumor detection with privacy preservation considerations for IoMT applications. LeNET achieved the highest accuracy at 98.7%.

B. Segmentation Architectures

Brain tumor segmentation demands pixel-level precision. Rajendran *et al.* [45] proposed combining U-Net and 3D CNN predictions through various merging strategies, using GLCM feature extraction for preprocessing. The combined approach achieved 99.40% mean accuracy and 99.41% precision on the BraTS dataset.

Tiwary *et al.* [50] replaced UNet's encoder with EfficientNet, initially frozen to retain pre-trained features, while preserving UNet's bottleneck and skip connections. On the Brain-Tumor.npy dataset (3,064 T1-weighted images from 233 patients), the model achieved 0.9925 accuracy.

Hussain *et al.* [51] compared UNet, Attention-UNet, and Residual-Attention-UNet for multi-task brain tumor segmentation and classification. The Residual-Attention-UNet achieved

TABLE II
SUMMARY OF SELECTED STUDIES IN PNEUMONIA DETECTION

Study	Architecture / Method	Dataset (Size)	Key Metric (%)
Sharma & Guleria [27]	VGG16 + NN	CXR Datasets	Acc: 95.4
Szepesi & Szilágyi [28]	Custom CNN with Dropout	Pediatric CXR (5,856)	Acc: 97.2
Shilpa *et al.* [31]	EfficientNetB0 + CLAHE	CXR Dataset	Acc: 99.78
Munir *et al.* [35]	PneuX-Net Ensemble	CXR Dataset	Acc: 99.91
DAViT [37]	ViT + CNN	Real-world CXR (5,856)	F1: 97.0
Li & Li [7]	CNN Voting Ensemble	COVID-19 CXR (10,182)	Acc: 99.95

89.30% Jaccard Similarity Index, 91.10% Dice Coefficient, and 93.35% overall accuracy. In binary classification, it reached 98.60% precision, 98.06% recall, and 99.40% accuracy.

The ZNet model [52] achieved a Dice similarity coefficient of 0.96 during training and 0.92 on independent testing, with pixel accuracy of 0.996, demonstrating the potential for automated brain tumor segmentation in 2D MRI.

Bagyaraj *et al.* [53] introduced U-Net-based dense convolutional networks for brain tumor cell segmentation, achieving sensitivity of 88.7%, Jaccard index of 0.839, dice score of 0.911, and specificity of 100% on a custom database of 300 HGG cases and 200 normal cases.

Zaitoon and Syed [54] developed the RU-Net2+ framework combining tumor segmentation with survival rate prediction. Using CNMF preprocessing and DBT-CNN classification on the BraTS dataset, they achieved 99.51% accuracy for HGG and 99.28% for LGG classification, with survival prediction accuracy of 85.71% (long-term), 72.72% (medium-term), and 61.54% (short-term).

C. YOLO-Based Detection Methods

Almufareh *et al.* [44] compared YOLOv5 and YOLOv7 for brain tumor detection and segmentation across three classes. YOLOv5 achieved mAP of 0.947 at IoU 0.5 with precision of 0.94, while YOLOv7 achieved 0.94 mAP with precision of 0.936. Both frameworks were systematically compared with RCNN, Faster RCNN, and Mask RCNN.

D. Ensemble and Multi-Modal Approaches

Singh *et al.* [55] introduced an ensemble model combining ResNet50 (97.65% validation accuracy) and EfficientNet-B7 (98.20%) for brain tumor classification on over 22,000 high-resolution MRI images. The ensemble achieved 99.68% validation accuracy and 99.53% post-training accuracy—exceeding both individual models.

Shyamala and Basha [56] proposed a multi-modal framework using both MRI and CT images with AlexNet, GoogleNet, SqueezeNet, and ResNet for feature extraction. Classifier-level fusion with neural networks outperformed SVM and KNN, with SqueezeNet-ResNet fusion achieving near-perfect classification.

Ahmmed *et al.* [57] developed BTdiagAI, integrating fine-tuned InceptionV3, DenseNet121, and Xception with ANOVA and L1max feature selection. Their systematic evaluation of five preprocessing methods identified CLAHE as optimal (99.8% accuracy on Dataset 1, 99.61% on Dataset 2), deploying results through a web platform for clinical use.

Majib *et al.* [58] proposed VGG-SCNet, a stacked classifier combining multiple state-of-the-art transfer learning models, achieving 99.2% precision, 99.1% recall, and 99.2% F1-score for brain tumor detection.

E. Review Papers and Surveys

Solanki *et al.* [2] provided a comprehensive overview of computational intelligence and statistical image processing techniques for brain tumor detection, comparing classical ML approaches with modern deep learning methods.

Karim *et al.* [3] investigated the landscape of brain tumor segmentation using emerging deep learning methods, offering comprehensive comparison while highlighting limitations related to data scarcity and validation robustness.

Jia and Chen [59] proposed FAHS-SVM using probabilistic neural network classification for brain tumor detection, achieving 98.51% accuracy in distinguishing normal from abnormal tissue in brain MRI.

F. Trends and Gaps in Brain Tumor Research

The brain tumor domain exhibits: (1) convergence toward U-Net-based segmentation architectures, with attention mechanisms and residual connections as standard enhancements; (2) ensemble methods combining multiple pre-trained architectures consistently outperforming individual models; (3) growing interest in multi-task frameworks that jointly perform classification, segmentation, and survival prediction. Key gaps include: reliance on the Figshare/BraTS benchmark datasets limiting generalizability; insufficient exploration of 3D volumetric analysis compared to 2D slice-based approaches; limited work on real-time intra-operative applications; and absence of multi-institutional prospective validation studies.

VI. CROSS-DOMAIN COMPARISON

A. Imaging Modalities and Input Data

The three domains employ fundamentally different imaging modalities. Pneumonia detection primarily uses chest X-rays (2D grayscale projections), with a small subset exploring CT. Brain tumor analysis uses MRI (multi-modal 3D volumes, though most studies analyze 2D slices). Gastric ulcer research uses endoscopic images (2D color images from direct visualization) and occasionally histopathological slides. Despite these differences, all three domains converge on similar deep

TABLE III
SUMMARY OF SELECTED STUDIES IN BRAIN TUMOR MRI ANALYSIS

Study	Architecture / Method	Dataset (Size)	Key Metric (%)
Noreen *et al.* [6]	Inception-v3 + DenseNet201	Public 3-class	Acc: 99.51
Rizwan *et al.* [48]	Gaussian CNN	Benchmark (3,064)	Acc: 99.8
Rajendran *et al.* [45]	U-Net + 3D CNN	BraTS	Acc: 99.40
Hussain *et al.* [51]	Residual-Attention-UNet	Brain MRI	Acc: 99.40
Zaitoon & Syed [54]	RU-Net2+	BraTS	Acc: 99.51
Singh *et al.* [55]	ResNet50 + EfficientNet-B7	High-Res MRI (22,000+)	Acc: 99.68

TABLE IV
DOMINANT DL ARCHITECTURES ACROSS DOMAINS

Architecture	Gastric	Pneumonia	Brain Tumor
ResNet variants	✓	✓	✓
VGG variants	–	✓	✓
DenseNet	✓	✓	✓
EfficientNet	–	✓	✓
U-Net variants	✓	–	✓
YOLO variants	✓	–	✓
Inception/Xception	✓	✓	✓
Vision Transformers	–	✓	–

learning backbones—particularly ResNet, VGG, DenseNet, and EfficientNet—suggesting these architectures offer sufficient feature extraction capacity across diverse medical imaging contexts.

B. Dominant Architectures and Methodological Trends

TABLE IV summarizes the dominant architectures across domains. ResNet variants (ResNet50, ResNet152) appear across all three domains as either primary classifiers or feature extractors. VGG architectures are predominantly used in pneumonia detection. U-Net and its variants are almost exclusively applied to brain tumor segmentation, though U-Net-based segmentation also appears in gastric research for lesion localization. YOLO-based object detection appears in both gastric (YOLOv7, YOLOv8) and brain tumor (YOLOv5, YOLOv7) domains. Vision Transformers have so far appeared primarily in pneumonia detection (DAViT, Swin Transformer), representing an emerging trend that may spread to other domains.

C. Evaluation Metrics and Validation Strategies

Accuracy, precision, recall, F1-score, and AUC-ROC are universally reported across all three domains. Brain tumor segmentation additionally employs Dice coefficient and Jaccard index. A concerning trend across all domains is the predominance of single-dataset evaluation without external validation. Cross-validation (typically 5-fold or 10-fold) is more common in pneumonia research, while brain tumor and gastric studies more frequently use train/validation/test splits.

D. Common Limitations

Several limitations recur across all three domains: (1) *Dataset homogeneity*: most studies use a single public dataset (Kermany for pneumonia, Figshare/BraTS for brain tumors, custom single-institution datasets for gastric ulcers); (2) *Class imbalance*: malignant cases are typically underrepresented relative to benign or normal classes; (3) *Absence of clinical deployment*: very few studies report prospective clinical trials or real-world deployment metrics; (4) *Limited demographic diversity*: datasets are often drawn from single geographic regions; and (5) *Reproducibility concerns*: many studies do not release code or trained models.

VII. RESEARCH GAPS

A. Gastric Ulcer Domain

- Video-based analysis of full endoscopic procedures remains underexplored compared to single-frame classification.
- Integration of non-imaging biomarkers (microbiota profiles [26], serology) with AI image analysis is absent.
- Real-time, prospective clinical studies comparing AI-assisted versus standard endoscopy have not been reported.
- Multi-institutional dataset standardization for benchmarking is lacking.

B. Pneumonia Domain

- Over-reliance on the Kermany pediatric dataset limits generalizability to adult populations and diverse imaging systems [41].
- Differentiation between viral subtypes (influenza, RSV, COVID-19) from CXR alone remains a substantial challenge.
- Federated learning approaches [39] require further maturation and large-scale multicenter validation.
- Integration of clinical metadata (patient age, symptoms, lab results) with image-based classification is minimal.

C. Brain Tumor MRI Domain

- True 3D volumetric segmentation is underrepresented; most studies analyze individual 2D MRI slices.
- Real-time intra-operative tumor delineation for surgical navigation has received limited attention.
- Survival prediction [54] is explored in very few studies despite its clinical significance.
- Multi-modal integration (combining MRI with CT, PET, or clinical data) remains nascent [56].

D. Cross-Domain Gaps

- Explainability techniques (Grad-CAM, SHAP, attention maps) are applied inconsistently across domains.
- Standardized reporting frameworks for AI medical imaging studies (analogous to CONSORT for clinical trials) are needed.
- Transfer of architectural innovations between domains (e.g., applying gastric endoscopy attention mechanisms to CXR analysis) is unexplored.
- Regulatory and ethical frameworks for clinical deployment of these AI systems remain underdeveloped.

VIII. Conclusion

This literature review has synthesized findings from sixty contemporary research papers across three critical domains of AI-assisted medical image analysis. While each domain presents unique challenges related to imaging modalities and diagnostic objectives, several overarching themes emerge regarding the maturity, limitations, and future trajectory of the field.

A. Gastric Ulcer: Key Takeaways

The gastric ulcer domain has progressed from basic binary (benign/malignant) classification to comprehensive diagnostic systems incorporating detection, localization, multi-class classification, and staging estimation. Dominant methods include CNN-based classification (ResNet, Xception, DenseNet), object detection frameworks (YOLOv7, YOLOv8), and attention mechanisms for focusing on diagnostically relevant regions. The field is approaching clinical readiness for specific tasks—malignancy differentiation accuracy exceeds 96% in controlled studies [5], [15]—but remains immature in terms of prospective clinical validation, multicenter generalizability, and integration with clinical workflows. Domain-specific pre-training (Gastro-BaseNet [19]) represents a promising direction that may generalize to other medical imaging domains.

B. Pneumonia: Key Takeaways

Pneumonia detection from CXR represents perhaps the most extensively studied of the three domains, driven by the availability of large public datasets and the urgency of the COVID-19 pandemic. Transfer learning with architectures such as EfficientNet, VGG, and ResNet achieves consistently high accuracy (94–99.9%), with ensemble and hybrid approaches reaching the performance ceiling on existing benchmarks. The field is transitioning toward: (a) Vision Transformer-based architectures offering global context [37], [38]; (b) explainable AI frameworks [40] providing clinical transparency; and (c) privacy-preserving federated learning [39]. However, the field's maturity is tempered by the critical observation that most high-accuracy results derive from a single dataset (Kermany), and performance on diverse real-world populations remains unvalidated [1], [41].

C. Brain Tumor MRI: Key Takeaways

Brain tumor MRI analysis has achieved exceptional classification accuracy (exceeding 99% on standard benchmarks) through ensemble transfer learning and dedicated segmentation architectures. U-Net variants enhanced with attention mechanisms and EfficientNet encoders represent the state of the art for segmentation, while ensemble models combining complementary architectures (ResNet50 + EfficientNet-B7 [55]) define the classification frontier. The domain exhibits the greatest methodological depth, with studies addressing classification, segmentation, detection, and survival prediction within unified frameworks [54]. The primary maturity bottleneck is the transition from benchmark performance to clinical deployment—the gap between 99%+ accuracy on curated datasets and reliable performance in diverse real-world clinical settings remains substantial. Multi-modal integration [56] and web-deployed diagnostic tools [57] represent emerging efforts to bridge this gap.

Acknowledgment

The authors would like to thank their faculty advisors and the Department of Computer Engineering at SVKM's NMIMS for their guidance and support throughout this capstone project.

References

[1] W. Khan, N. Zaki, and L. Ali, "Intelligent pneumonia identification from chest X-rays: A systematic literature review," *IEEE Access*, vol. 9, pp. 51747–51771, Mar. 2021.

[2] S. Solanki, U. P. Singh, S. S. Chouhan, and S. Jain, "Brain tumor detection and classification using intelligence techniques: An overview," *IEEE Access*, vol. 11, pp. 12870–12886, Feb. 2023.

[3] S. Karim *et al.*, "Developments in brain tumor segmentation using MRI: Deep learning insights and future perspectives," *IEEE Access*, vol. 12, pp. 25958–25983, Feb. 2024.

[4] J.-Y. Sun, S.-W. Lee, M.-C. Kang, S.-W. Kim, S.-Y. Kim, and S.-J. Ko, "A novel gastric ulcer differentiation system using convolutional neural networks," in *Proc. IEEE 31st Int. Symp. Comput.-Based Med. Syst. (CBMS)*, 2018, pp. 351–356.

[5] M. Sun, K. Liang, W. Zhang, Q. Chang, and X. Zhou, "Non-local attention and densely-connected convolutional neural networks for malignancy suspiciousness classification of gastric ulcer," *IEEE Access*, vol. 8, pp. 15812–15822, Jan. 2020.

[6] N. Noreen, S. Palaniappan, A. Qayyum, I. Ahmad, M. Imran, and M. Shoaib, "A deep learning model based on concatenation approach for the diagnosis of brain tumor," *IEEE Access*, vol. 8, pp. 55135–55144, Mar. 2020.

[7] D. Li and S. Li, "An artificial intelligence deep learning platform achieves high diagnostic accuracy for Covid-19 pneumonia by reading chest X-ray images," *iScience*, vol. 25, no. 4, p. 104031, Apr. 2022.

[8] H. Aljuaid, H. Adlan, B. Alkebsi, B. S. Alfurhood, A. Liotta, and L. Cavallaro, "An experimental comparison of deep learning models for pneumonia classification from chest X-ray images," *Biomed. Signal Process. Control*, 2026.

[9] S.-A. Lee, H. C. Cho, and H.-C. Cho, "A novel approach for increased convolutional neural network performance in gastric-cancer classification using endoscopic images," *IEEE Access*, vol. 9, pp. 51847–51854, Mar. 2021.

[10] S. Hatami, M. H. Olyaei, and R. Shamsaee, "Detection and classification of gastric precancerous diseases using deep learning," in *Proc. 6th Iranian Conf. Signal Process. Intell. Syst. (ICSPIS)*, 2020, pp. 1–5.

[11] H. Olimboyev *et al.*, "Development of the Gastro-AI model for diagnosing gastrointestinal diseases based on the HyperKvasir," in *Proc. IEEE Conf.*, 2024.

[12] T. A. Kuchkorov, N. Q. Sabitova, and T. D. Ochilov, "Detection of gastric ulcers and lesions applying CNN architecture," in *Proc. Int. Sci. Tech. Conf. Comput. Sci. Eng. Technol.*, 2022, pp. 200–205.

[13] S. Budhiraja and N. Batra, "Deep learning based approaches for gastric cancer detection and lesion segmentation in gastrointestinal endoscopy images," in *Proc. IEEE Conf.*, 2023.

[14] S. Ahmad, J.-S. Kim, D. K. Park, and T. Whangbo, "Automated detection of gastric lesions in endoscopic images by leveraging attention-based YOLOv7," *IEEE Access*, vol. 11, pp. 73418–73429, Jul. 2023.

[15] Y. Tan *et al.*, "Deep learning-based AI system for real-time differentiation of benign and malignant gastric ulcers," *BMC Gastroenterol.*, vol. 26, p. 378, 2026.

[16] B. S. Kim *et al.*, "Enhanced multi-class pathology lesion detection in gastric neoplasms using deep learning-based approach and validation," *Sci. Rep.*, vol. 14, p. 11527, 2024.

[17] X. Zheng, Z. Zeng, C. Ma, Q. Chang, Z. Zhao, and X. Yang, "Malignancy suspicious region guided deep neural networks for gastric ulcer classification," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2021.

[18] M. Ramamoorthy *et al.*, "Early cancer diagnosis and ulcer classifications in endoscopy videos using deep learning approach," in *Proc. IEEE Conf.*, 2023.

[19] G. P. Lee, Y. J. Kim, D. K. Park, Y. J. Kim, S. K. Han, and K. G. Kim, "Gastro-BaseNet: A specialized pre-trained model for enhanced gastroscopic data classification and diagnosis of gastric cancer and ulcer," *Diagnostics*, vol. 14, no. 1, p. 75, 2024.

[20] R. Yang *et al.*, "Artificial intelligence efficiently predicts gastric lesions, Helicobacter pylori infection and lymph node metastasis upon endoscopic images," *Chin. J. Cancer Res.*, vol. 36, no. 5, pp. 489–502, 2024.

[21] M. Zubair, M. Owais, T. Mahmood, S. Iqbal, S. M. Usman, and I. Hussain, "Enhanced gastric cancer classification and quantification interpretable framework using digital histopathology images," *Sci. Rep.*, vol. 14, 2024.

[22] A. Saba, J. Amin, and M. U. Ali, "Deep Q-learning for gastrointestinal disease detection and classification," *Bioengineering*, vol. 12, no. 11, p. 1184, 2025.

[23] Y. Masmoudi, M. Ramzan, S. A. Khan, and M. Habib, "Optimal feature extraction and ulcer classification from WCE image data using deep learning," *Soft Comput.*, vol. 26, pp. 7571–7583, 2022.

[24] S. Gnanapriya, D. Reshma, and M. Varun, "Identification of gastric abnormalities with synchronous biopsy site localization from endoscope imaging in gastro-intestinal tract using deep learning technique," in *Proc. IEEE Conf.*, 2023.

[25] H. J. Yoon and J.-H. Kim, "Lesion-based convolutional neural network in diagnosis of early gastric cancer," *Clin. Endosc.*, vol. 53, no. 2, pp. 127–131, 2020.

[26] N. Sari *et al.*, "Gastric microbiota composition in Turkish patients with non-ulcer dyspepsia and peptic ulcer disease," *BMC Microbiol.*, vol. 26, p. 46, 2026.

[27] S. Sharma and K. Guleria, "A deep learning based model for the detection of pneumonia from chest X-ray images using VGG-16 and neural networks," *Procedia Comput. Sci.*, vol. 218, pp. 357–366, 2023.

[28] P. Szepesi and L. Szilágyi, "Detection of pneumonia using convolutional neural networks and deep learning," *Biocybern. Biomed. Eng.*, vol. 42, no. 3, pp. 1012–1022, 2022.

[29] S. Potharaju, S. N. Tambe, K. Dasari, N. Srikanth, R. Venkatarao, and S. Tambe, "Enhanced X-ray image classification for pneumonia detection using deep learning based CBAM and SE mechanisms," *Intell.-Based Med.*, 2025.

[30] M. Ali *et al.*, "Pneumonia detection using chest radiographs with novel EfficientNetV2L model," *IEEE Access*, vol. 12, pp. 24456–24467, Mar. 2024.

[31] N. Shilpa, W. A. Banu, and P. B. Metre, "Revolutionizing pneumonia diagnosis: AI-driven deep learning framework for automated detection from chest X-rays," *IEEE Access*, vol. 12, pp. 171601–171616, Nov. 2024.

[32] H. Lamouadene *et al.*, "Detection of COVID-19, lung opacity, and viral pneumonia via X-ray using machine learning and deep learning," *Comput. Biol. Med.*, 2025.

[33] P. Agrawal and S. Gandhi, "PneumoSense++: An enhanced pneumonia detection with deep learning and comparative optimizer analysis," *Procedia Comput. Sci.*, vol. 283, pp. 3407–3423, 2026.

[34] S. Akshay, S. Kashyap, H. N. D. Patel, and Y. R. P. Kumar, "Hybrid deep learning and machine learning framework for automated pneumonia detection in chest X-ray images," *MethodsX*, vol. 15, p. 103729, 2025.

[35] K. Munir, M. U. Tanveer, H. J. Alyamani, A. Bermak, and A. U. Rehman, "PneuX-Net: An enhanced feature extraction and transformation approach for pneumonia detection in X-ray images," *IEEE Access*, vol. 13, May 2025.

[36] M. F. Hashmi, A. A. Husain, D. M. Kotambkar, and K. D. Kumar, "Hybrid deep learning and handcrafted feature fusion for pneumonia detection in chest X-rays," *J. Electr. Syst. Inf. Technol.*, vol. 12, p. 94, 2025.

[37] M. Fu, C. Tantithamthavorn, and T. Le, "DAViT: A domain-adapted vision transformer for automated pneumonia detection and explanation using chest X-ray images," *IEEE Access*, 2025.

[38] A. Kunjumon, S. V, and J. J. Nair, "A hybrid deep learning framework for improved pneumonia detection in chest X-ray images," *Procedia Comput. Sci.*, vol. 282, pp. 1043–1055, 2026.

[39] N. Rana and H. Marwaha, "Pneumonia detection from X-ray images using federated learning—an unsupervised learning approach," *Meas.: Sensors*, 2025.

[40] H. Ren *et al.*, "Interpretable pneumonia detection by combining deep learning and explainable models with multisource data," *IEEE Access*, vol. 9, pp. 95523–95533, Jun. 2021.

[41] D. Rickard, M. A. Kabir, and N. Homaira, "Machine learning-based approaches for distinguishing viral and bacterial pneumonia in paediatrics: A scoping review," *Comput. Methods Programs Biomed.*, 2025.

[42] Md. R. Hasan, S. M. A. Ullah, and S. Md. R. Islam, "Recent advancement of deep learning techniques for pneumonia prediction from chest X-ray image," *Medical Reports*, vol. 7, p. 100106, 2024.

[43] A. Younis *et al.*, "Abnormal brain tumors classification using ResNet50 and its comprehensive evaluation," *IEEE Access*, vol. 12, pp. 80470–80480, May 2024.

[44] M. F. Almufareh, M. Imran, A. Khan, M. Humayun, and M. Asim, "Automated brain tumor segmentation and classification in MRI using YOLO-based deep learning," *IEEE Access*, vol. 12, pp. 16189–16204, Jan. 2024.

[45] S. Rajendran *et al.*, "Automated segmentation of brain tumor MRI images using deep learning," *IEEE Access*, vol. 11, pp. 64758–64768, Jun. 2023.

[46] S. Asif, W. Yi, Q. U. Ain, J. Hou, T. Yi, and J. Si, "Improving effectiveness of different deep transfer learning-based models for detecting brain tumors from MR images," *IEEE Access*, vol. 10, pp. 34716–34730, Feb. 2022.

[47] S. Ahmad and P. K. Choudhury, "On the performance of deep transfer learning networks for brain tumor detection using MR images," *IEEE Access*, vol. 10, pp. 59541–59556, May 2022.

[48] M. Rizwan, A. Shabbir, A. R. Javed, M. Shabbir, T. Baker, and D. Al-Jumeily Obe, "Brain tumor and glioma grade classification using Gaussian convolutional neural network," *IEEE Access*, vol. 10, pp. 29731–29740, Feb. 2022.

[49] M. U. Rehman *et al.*, "An efficient deep learning model for brain tumour detection with privacy preservation," *CAAI Trans. Intell. Technol.*, 2023.

[50] P. K. Tiwary, P. Johri, A. Katiyar, and M. K. Chhipa, "Deep learning-based MRI brain tumor segmentation with EfficientNet-enhanced UNet," *IEEE Access*, vol. 13, pp. 49651–49667, Mar. 2025.

[51] S. S. Hussain, N. A. Wani, J. Kaur, N. Ahmad, and S. Ahmad, "Next-generation automation in neuro-oncology: Advanced neural networks for MRI-based brain tumor segmentation and classification," *IEEE Access*, vol. 13, pp. 37483–37500, Mar. 2025.

[52] M. A. Ottom, H. A. Rahman, and I. D. Dinov, "Znet: Deep learning approach for 2D MRI brain tumor segmentation," *IEEE J. Transl. Eng. Health Med.*, vol. 10, pp. 1–10, 2022.

[53] S. Bagyaraj, R. Tamilselvi, P. B. M. Gani, and D. Sabarinathan, "Brain tumour cell segmentation and detection using deep learning networks," *IET Image Process.*, vol. 15, no. 10, pp. 2305–2315, 2021.

[54] R. Zaitoon and H. Syed, "RU-Net2+: A deep learning algorithm for accurate brain tumor segmentation and survival rate prediction," *IEEE Access*, vol. 11, pp. 112626–112643, Oct. 2023.

[55] R. Singh, S. Gupta, S. Bharany, A. Almogren, A. Altameem, and A. U. Rehman, "Ensemble deep learning models for enhanced brain tumor classification by leveraging ResNet50 and EfficientNet-B7 on high-resolution MRI images," *IEEE Access*, vol. 12, pp. 171862–171880, Nov. 2024.

[56] N. Shyamala and S. M. Basha, "Multi-modal deep feature extraction and classifier-level integration for brain tumour classification using CT and MRI image," *Artif. Intell. Eng.*, 2026.

[57] F. Ahmmed *et al.*, "BTdiagAI: A web-deployed hybrid framework for brain tumor classification using optimized MRI preprocessing and deep learning fusion," *Healthcare Technol. Lett.*, 2025.
[58] M. S. Majib, M. M. Rahman, T. M. S. Sazzad, N. I. Khan, and S. K. Dey, "VGG-SCNet: A VGG Net-based deep learning framework for brain tumor detection on MRI images," *IEEE Access*, vol. 9, pp. 116942–116952, Aug. 2021.
[59] Z. Jia and D. Chen, "Brain tumor identification and classification of MRI images using deep learning techniques," *IEEE Access*, vol. 8, pp. 123783–123797, Aug. 2020.